

Drone Delivery Feasibility Study

Taclo — Hybrid Drone + Rider Model | Bangalore Pilot | India Food-Tech Platform

Model Scope: Unit Economics · 5-Year P&L · Sensitivity · Scenario Analysis · Break-Even

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KEY FACTS

₹45

Drone cost/order (base)

₹67

Rider cost/order

404

Break-even orders/drone/month

~30.6%

EBITDA margin (Year 1)

₹103L

5-Yr NPV (Base Case)

WHAT THIS STUDY ANSWERS

Can drones beat riders on cost?

What order volume is needed to break even?

What does a 5-year P&L look like?

Which scenario (Bear/Base/Bull) is investable?

Where does Taclo go first — and how fast?

THE IDEA

Taclo – To adapt drone delivery or not?

₹67/order

Current rider delivery cost, with 10% monthly attrition adding hidden cost

500K orders/mo

Bangalore platform volume — only 30% in drone-serviceable radius

INR 21.9 savings

Theoretical cost advantage of drones at base utilisation (600 ord/mo)

DGCA sandbox

Regulatory environment is evolving — first mover gets the window

Unit Economics: Rider vs. Drone

PER-ORDER COST BREAKDOWN — WHAT DOES ONE DELIVERY ACTUALLY COST?

DRONE SAVES ₹21.9 PER ORDER			
RIDER MODEL — ₹66.9 / order			
Fixed salary	₹4,000/mo	÷ 425 orders	₹9.41
Delivery bonus	₹55/order	piece-rate	₹55.00
Attrition cost	₹1,062/mo	10% churn × ₹1,500	₹2.50
Fuel & vehicle	Rider-borne	—	₹0
TOTAL RIDER COST / ORDER			₹66.91
DRONE MODEL — ₹45.0 / order (Base)			
Capex amortisation	₹130K ÷ 21 mo		₹6,190/mo
Battery replacement	₹13K ÷ 3 mo		₹4,333/mo
Maintenance	Case study		₹6,500/mo
Insurance + Ops + Hub	Blended overhead		₹10,000/mo
TOTAL MONTHLY COST / DRONE			₹27,024

₹90.1

Conservative
300 ord/mo

₹45.0

Base Case
600 ord/mo

₹36.0

Aggressive
750 ord/mo

Break-Even Analysis

AT WHAT VOLUME DO DRONES BEAT RIDERS? HOW FAST DOES THE FLEET PAY BACK?

404

orders / drone / month

Break-Even Volume

₹27,024

monthly cost / drone

Total Fixed Overhead

67%

capacity utilisation

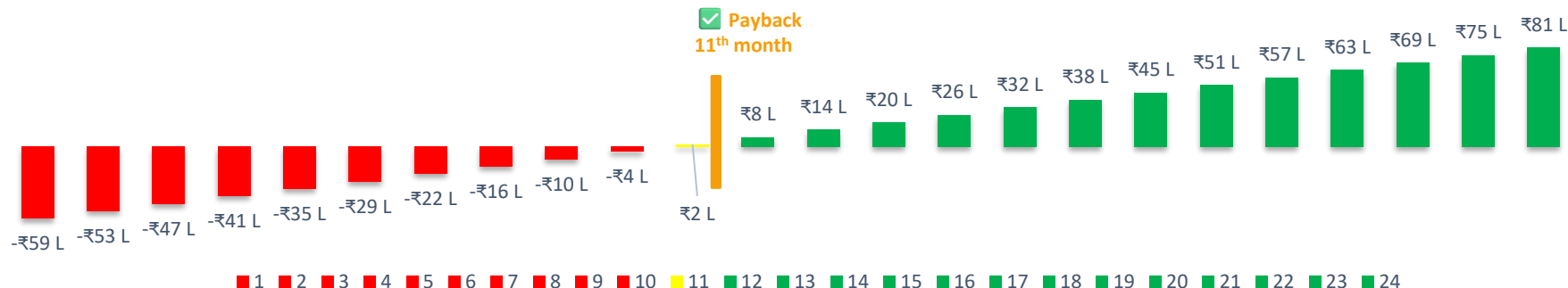
At Break-Even Point

Month 11

base case

Fleet Payback Period

PAYBACK TIMELINE (50-drone fleet, base case — 600 orders/drone/month)

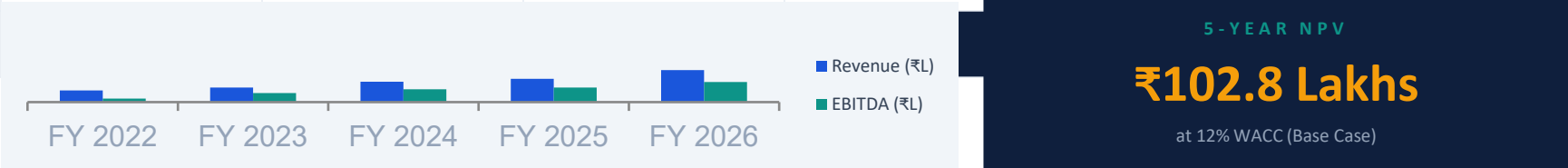


⚠️ Conservative case (480 ord/drone/mo): break-even at ~month 24 — fleet still viable but requires stronger demand assumptions before City 2 expansion, Hence only to deploy in busy areas

5-Year Financial Projections

REVENUE, EBITDA & FREE CASH FLOW — BASE CASE (20→55 DRONE FLEET SCALE)

Metric (₹ Lakhs)	FY 2022	FY 2023	FY 2024	FY 2025	FY 2026
Active Drones (units)	20	25	35	40	55
Total Drone Orders/Yr	1.44L	1.80L	2.52L	2.88L	3.96L
Total Revenue	79.2	99.0	138.6	158.4	217.8
Total OPEX	55.0	37.0	51.5	59.5	80.5
EBITDA	24.2	62.0	87.1	98.9	137.3
EBITDA Margin	30.6%	62.6%	62.8%	62.4%	63.0%
Net Profit	7.0	10.1	14.3	15.9	22.8
Free Cash Flow	-4.1	26.1	36.8	41.6	58.2



Sensitivity Analysis

HOW DOES COST/ORDER CHANGE AS MONTHLY DRONE COST AND ORDER VOLUME VARY?

Mon orders/→ ₹ Drone monthly exp ↓	300	400	500	600	700	800
₹25K	₹83.3	₹62.5	₹50.0	₹41.7	₹35.7	₹31.3
₹26K	₹86.7	₹65.0	₹52.0	₹43.3	₹37.1	₹32.5
₹27K	₹90.0	₹67.5	₹54.0	₹45.0	₹38.6	₹33.8
₹28K	₹93.3	₹70.0	₹56.0	₹46.7	₹40.0	₹35.0
₹29K	₹96.7	₹72.5	₹58.0	₹48.3	₹41.4	₹36.3
₹30K	₹100.0	₹75.0	₹60.0	₹50.0	₹42.9	₹37.5
₹31K	₹103.3	₹77.5	₹62.0	₹51.7	₹44.3	₹38.8
₹32K	₹106.7	₹80.0	₹64.0	₹53.3	₹45.7	₹40.0

> ₹67 (unviable — worse than rider)

₹45–₹67 (below rider)

< ₹45 (excellent)

KEY INSIGHTS

500+ orders

minimum monthly volume for drone to match rider cost at current overhead

Only 4 cells

in the entire matrix beat rider on cost — all need 600+ orders/mo

₹27K overhead

is the key lever — reducing it by 10% expands the green zone significantly

Base case (₹27K, 600 ord)

gives ₹45/order — 33% cheaper than rider

Scenario Analysis: Bear / Base / Bull for YEAR 3

ALL OUTPUTS FULLY FORMULA-DRIVEN — NO HARDCODED NUMBERS IN THE MODEL



BEAR CASE

KEY INPUTS

- 400 orders/drone/mo
- 20 drones — Year 3
- 60% pure drone mix
- ₹40 revenue/order
- WACC 15%
- 12-mo regulatory delay

OUTPUTS

Year 3 Trips/mo	240/mo
Year 3 Revenue	₹1.01L
Year 3 OPEX	₹1.44L
Year 3 EBITDA	-₹0.43L
Cost/Delivery	₹30

HOLD



BASE CASE

KEY INPUTS

- 550 orders/drone/mo
- 30 drones — Year 3
- 70% pure drone mix
- ₹55 revenue/order
- WACC 12%
- 3-mo regulatory delay

OUTPUTS

Year 3 Trips/mo	385/mo
Year 3 Revenue	₹2.28L
Year 3 OPEX	₹1.20L
Year 3 EBITDA	₹1.08L
Cost/Delivery	₹18.2

PROCEED



BULL CASE

KEY INPUTS

- 700 orders/drone/mo
- 40 drones — Year 3
- 80% pure drone mix
- ₹65 revenue/order
- WACC 9%
- 0-mo regulatory delay

OUTPUTS

Year 3 Trips/mo	560/mo
Year 3 Revenue	₹4.09L
Year 3 OPEX	₹0.96L
Year 3 EBITDA	₹3.13L
Cost/Delivery	₹11.4

ACCELERATE

* Hybrid orders = drone carries to proximity hub + rider handles last-mile (e.g. high-rise buildings, narrow lanes)

Phased City Rollout Plan

City-by-city expansion — regulatory-led, economics-validated



3-YEAR CAPEX SUMMARY

₹45.5L	₹32.5L	₹26L	₹19.9L	₹15.5L	₹7.25L	₹284.75L
BLR Fleet	HYD Fleet	Delhi/Mum	Hub Infra	Tech & SW	Training	TOTAL 3-YR

Decision Matrix: Go / No-Go Scorecard

WEIGHTED CRITERIA SCORING — BEAR / BASE / BULL — WITH RISK REGISTER

Criteria	Weight	 Bear	 Base	 Bull	Wtd Score (Base)
Financial ROI & Payback	25%	2	4	5	1.0
Operational Feasibility	20%	3	4	4	0.8
Scalability Potential	15%	2	4	5	0.6
Regulatory Environment	12%	2	3	4	0.36
Technology Readiness	10%	3	4	5	0.4
Competitive Differentiation	10%	3	4	5	0.4
Customer Experience	5%	4	4	5	0.2
ESG / Rider Impact	3%	2	3	3	0.09
TOTAL WEIGHTED SCORE	100%	2.50	3.85	4.62	3.85

TOP RISKS			
 DGCA delay	 Rider backlash	 Weather ops	 Low adoption
File sandbox early	Phased + retraining	AI routing algo	Free first delivery

VERDICT



Phased Rollout Recommended

< 2.5 → No-Go

2.5–3.5 → Conditional

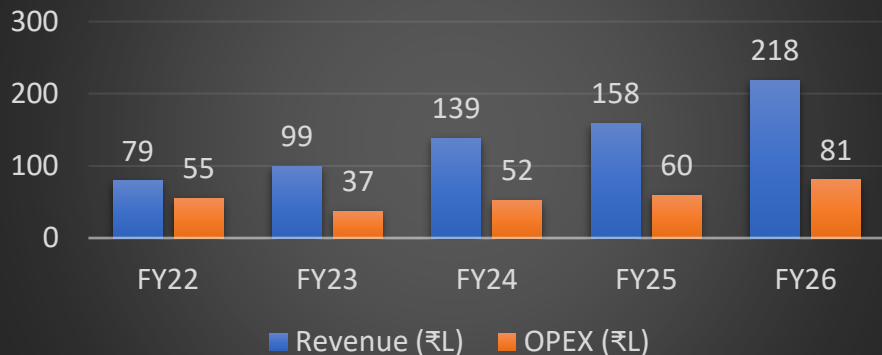
> 3.5 → Go 

Base score: 3.85
→ Proceed with Phase 1 Pilot

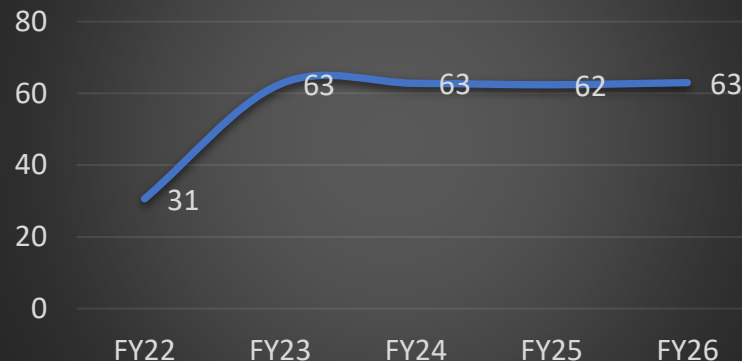
Financial Deep Dive: Revenue, EBITDA & FCF

5-YEAR TRAJECTORY — FLEET GROWS FROM 20 TO 55 DRONES ACROSS BANGALORE

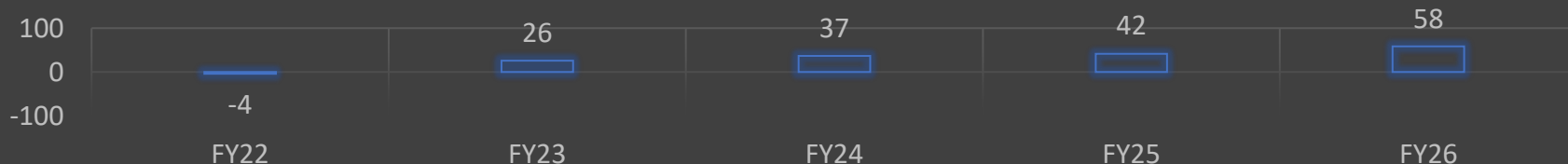
Revenue vs OPEX (₹ Lakhs)



EBITDA Margin (%)



Annual Free Cash Flow (₹ Lakhs) — Positive from Year 2





Proceed with Phased Rollout

The base case is financially sound: drones save ₹21.9 per order vs riders, break even at 404 orders/month, pay back the fleet in 11 months, and deliver a 5-year NPV of ₹102.8 Lakhs at 12% WACC.

Immediate (0–3 mo)

- › File DGCA sandbox application
- › Deploy 5-drone BLR pilot in dense zip codes
- › Build demand forecasting model for dispatch

Short-Term (3–9 mo)

- › Scale BLR to 25 drones if pilot hits 400+ ord/mo
- › Launch rider retraining program
- › Finalise hub-and-spoke infrastructure

Medium-Term (9–18 mo)

- › Activate Hyderabad with 20 drones
- › Implement live drone tracking in-app
- › Negotiate battery-as-a-service contracts

Watch Closely

- › If orders/drone/mo < 350 in BLR for 3 months
- › Reassess expansion to city 2
- › Consider variable pricing model to drive volume

Appendix: Model Assumptions & Sources

ALL INPUTS WITH SOURCE ATTRIBUTION — FULL MODEL AVAILABLE IN EXCEL

RIDER MODEL

Monthly salary	₹4,000
Bonus/delivery	₹55
Avg deliveries/mo	425
Attrition rate	10%/month
Recruit cost	₹1,500

DRONE MODEL

Drone cost	₹1.3 Lakh
Useful life	21 months
Maintenance/mo	₹6,500
Battery cost	₹13,000 per unit
Battery life	3 months
Overhead/mo	₹10,000
Avg speed	40 km/h
Flight range	7.5 km

MARKET

BLR platform orders/mo	500,000
Drone-serviceable zone	30%
Year 1 capture rate	5%
Year 3 capture rate	15%
Delivery fee	₹40/order
Drone premium	₹15/order
Market CAGR	27.5%

FINANCIAL

WACC	12%
Corporate tax	25%
Fleet: FY22-FY26	20/25/35/40/55
NPV horizon	5 years

Source: Case study + Analyst assumptions